

Technical Progress Sheet

1. Project Summary & Problem Addressed

Aging water supply infrastructure suffers from unmonitored pipe blockages, leakages, and delayed responses to pump failures. Neo Net is an IoT-enabled smart water management solution that uses multi-sensor nodes, edge-level switching, and real-time telemetry to automate pump operations and detect pipeline anomalies dynamically.

2. Key Achievements & Proof-of-Concept

Sensor Calibration: Calibrated turbidity (voltage-to-NTU mapping), ultrasonic (water level thresholding), and flow sensors (pulse counting).

Hardware Integration: Implemented a high-efficiency MOSFET driver module circuit to control a 12V water pump safely with minimal heat dissipation.

Physical Loop Testing: Built a physical pipeline circulation network simulating real-world fluid flow, automatic pump activation/cutoff, and tank depth tracking.

Web Dashboard: Developed a user interface for live parameter monitoring (flow rate, turbidity, water level) and manual overrides.

3. Key Performance & Validation Metrics

Automatic Cutoff Speed: Pump power disconnects within ~180 ms when lower water level thresholds are reached, preventing dry running.

Sensor Stability: Turbidity sensor yields stable readings within $\pm 4\%$ accuracy in tested laboratory samples.

4. Identified Limitations & Immediate Next Steps

Wi-Fi Reliance: Cloud syncing currently depends on active Wi-Fi (local edge logic handles immediate safety cutoffs).

Next Steps for Finals: Design a custom PCB on Easy EDA, integrate edge machine learning for leak prediction, and add LoRa/Cellular fallback.